Nexys A7-100T / Artix-7 XC7A100T

Comprehensive Research Report for FPGA Video Game Development

Hardware resources | Sensors | GPIO & serial interfaces | Controllers | 2D/3D graphics | Memory | Clocking

Scope assumption - This report assumes the Digilent Nexys A7-100T board with XC7A100T-1CSG324C. If your board is Arty A7-100T or another XC7A100T board, the FPGA resource section still applies, but onboard I/O and sensors must be regenerated for that board.

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1. Executive summary

The Nexys A7-100T is a strong platform for an educational video game implemented entirely in FPGA logic. Its most useful game-facing resources are the RGB444 VGA output, 128 MiB DDR2, five directional buttons, 16 switches, ADXL362 3-axis accelerometer, PDM microphone, audio output, USB-UART, a limited keyboard/mouse USB-HID bridge, four Pmod headers and an XADC expansion header. The FPGA itself supplies 63,400 6-input LUTs, 126,800 flip-flops, 4,860 Kb of block RAM, 1,188 Kb of distributed RAM, 240 DSP48E1 slices and six clock-management tiles.

For a first playable result, target 640x480 VGA with a tile/sprite engine and fixed 60-Hz-class frame timing. Keep hot graphics data in BRAM and use DDR2 for large framebuffers, depth buffers and textures. A full 3D pipeline is feasible as a custom accelerator, but there is no hard GPU; fixed-point arithmetic, pipelining and careful memory arbitration are essential.

Two documentation discrepancies - (1) Older Digilent text labels 1,188 Kb as Block RAM; AMD DS180 shows this is distributed RAM and that true XC7A100T Block RAM is 4,860 Kb. (2) the reference manual names an ADMP421 microphone, while the Rev. D.0 schematic shows SPK0833LM4H. The board-level interface remains PDM, but confirm your PCB revision for microphone-specific parameters.

2. Hardware identification and primary sources

Item	Value / source	Why it matters
Board assumption	Digilent Nexys A7-100T	Determines onboard sensors, connectors, memory and pinout.
FPGA	XC7A100T-1CSG324C	Artix-7, speed grade -1, CSG324 package.
Board reference manual	Digilent Nexys A7 Reference Manual	Primary board features and interface behavior.
Board schematic	Nexys A7 Rev. D.0 schematic	Best source for actual component part numbers and wiring.
FPGA overview	AMD DS180	Authoritative resource counts/package combinations.
Electrical/timing	AMD DS181 + UG471	Device AC/DC limits and SelectIO behavior.
Pin constraints	Digilent Nexys-A7-100T-Master.xdc	Canonical board-to-FPGA pin mapping.

3. XC7A100T hardware limits

Resource	XC7A100T	Design implication
Logic cells	101,440	Family capacity metric; do not equate directly to LUT count.
CLB slices	15,850	Each 7-series slice contains 4 LUTs and 8 flip-flops.
6-input LUTs	63,400	Main combinational logic budget.
Flip-flops	126,800	Pipeline/state storage.

Resource	XC7A100T	Design implication
Distributed RAM	1,188 Kb max	Uses LUT fabric; useful for small tables/FIFOs.
Block RAM	4,860 Kb = 135 x 36 Kb	About 607.5 KiB total on-chip BRAM.
DSP48E1	240	Excellent for fixed-point transforms, filters, MACs.
Clock Management Tiles	6	Each contains MMCM + PLL; derive pixel/core clocks.
XADC	1	Dual 12-bit, up-to-1-MSPS ADC architecture.
CSG324 user I/O	210 max	Package limit, not all pins are broken out on Nexys connectors.
GTP in CSG324	0	No multi-gigabit transceiver links on this package.

The Nexys A7 product literature mentions internal clock capability above 450 MHz, but this is not a guaranteed system clock for an arbitrary game design. Achievable Fmax is determined by the implemented netlist, speed grade, placement/routing and timing constraints. Start conservative, pipeline long datapaths, and use post-route timing reports as the authority.

4. Board resources useful for a game

Subsystem	Board resource	Game relevance
Clock	100 MHz oscillator	Reference for MMCM/PLL-generated clocks.
External RAM	128 MiB DDR2, 16-bit	Framebuffers, textures, large maps/assets.
Nonvolatile	16 MiB QSPI + microSD	Configuration/assets/storage.
Video	12-bit VGA: RGB444 + HS/VS	4096 colors; FPGA generates timing/pixels.
Audio output	Mono PWM/PDM path	Music/SFX via 1-bit modulation.
Audio input	PDM microphone	Clap/rhythm/voice-energy control.
Motion	ADXL362 3-axis accelerometer	Tilt/shake control. No gyroscope onboard.
Temperature	ADT7420 I2C	Telemetry; less relevant as direct player input.
Human I/O	16 switches, 5 buttons + reset	D-pad/menu/debug/configuration.
USB HID	Boot keyboard OR mouse -> PS/2	Useful input, but not arbitrary USB gamepad.
Serial	USB-UART TX/RX, optional RTS/CTS	Debug console or PC-to-FPGA control bridge.

Subsystem	Board resource	Game relevance
Network	10/100 Ethernet RMII	Optional multiplayer/telemetry; more complex.
Expansion	4 Pmod + JXADC	External joystick, gyro/IMU, encoders, sensors.

5. Onboard sensors and input devices

5.1 ADXL362 3-axis accelerometer

The onboard ADXL362 is the most immediately useful non-button game sensor. It provides 12-bit acceleration data on X/Y/Z axes, selectable +/-2 g, +/-4 g and +/-8 g ranges, communicates over SPI, and has two interrupt outputs. A typical game control chain is: SPI sampling -> signed conversion -> offset calibration -> low-pass filter -> dead zone -> gain/curve -> normalized control axis.

- Use tilt angle or gravity vector for steering/balance games.
- Use high-pass/threshold logic for shake, tap-like or motion gestures.
- The sensor is mounted on the FPGA board, so motion controls require physically moving the board. For handheld controls, an external IMU/joystick is ergonomically better.

5.2 ADT7420 temperature sensor

The ADT7420 is a high-accuracy digital temperature sensor connected over I2C. The Nexys schematic straps its address to binary 1001011 (0x4B) and exposes INT/CT signals. It is useful for board telemetry, thermal display or unusual game mechanics, but it is not a normal high-rate player input.

5.3 PDM microphone

The FPGA supplies a microphone clock and receives a 1-bit PDM data stream. To turn it into a game control, implement a decimator/low-pass chain (often CIC + FIR or another suitable filter), then compute amplitude/envelope or frequency features. A simple clap detector can be much cheaper than full audio reconstruction.

Revision check - Rev. D.0 schematic shows SPK0833LM4H, while the reference manual text names ADMP421. Code against the PDM interface first and use your board revision/BOM for component-specific limits.

5.4 Buttons, switches and USB HID

The five general pushbuttons are normally low and go high when pressed; the CPU reset button is normally high and goes low when pressed. Mechanical inputs need debouncing. The 16 slide switches are static digital levels. The onboard USB host subsystem supports one boot-protocol HID keyboard OR mouse and presents it to the FPGA as a PS/2-like two-wire interface. A generic USB gamepad should not be assumed to work through this bridge.

6. GPIO and serial protocols

The key FPGA concept is that "UART", "SPI" and "I2C" are not fixed peripheral blocks attached to every external pin. You assign I/O pins and synthesize the protocol logic. Therefore, almost any moderate-speed digital protocol can be implemented if the electrical levels and timing are compatible.

Protocol / interface	How to use it	Typical game device	Important caveat
GPIO	Direct input/output RTL	Buttons, D-pad, encoder, LEDs	Pmod logic is 3.3 V; level-shift 5 V.
SPI	RTL controller or IP	ADXL362, Pmod JSTK2, IMUs, ADCs	Easy high-rate sensor/control bus; separate chip selects.

Protocol / interface	How to use it	Typical game device	Important caveat
I2C	Open-drain controller in RTL/IP	ADT7420, many IMUs	Requires pull-ups and correct 3.3 V electrical design.
UART	RTL/IP; onboard USB-UART or Pmod pins	PC bridge, MCU controller, BLE module	Asynchronous; simple but lower throughput than SPI.
PS/2	Onboard HID bridge	Keyboard or mouse	Bridge is limited; not generic USB gamepad.
PDM	Clock + 1-bit stream	Onboard microphone	Requires digital filtering/decimation.
PWM	Timer/modulator	Audio, LED, servo via proper interface	Do not drive motors/loads directly from FPGA pin.
SD / SPI	microSD connector	Asset storage	Filesystem parsing adds complexity.
RMII	Ethernet MAC + PHY interface	Networking	Significant protocol stack complexity.
Custom synchronous serial	User RTL	Custom controller/sensor	Good FPGA strength if both endpoints are controlled.

6.1 Pmod electrical and expansion limits

Each standard Pmod connector has eight signal positions plus 3.3 V and ground. Four Pmods expose 32 general digital signal positions. The Digilent master XDC uses LVCMOS33 on these headers. Treat these inputs as 3.3 V logic; do not connect 5 V output modules directly. The Pmod routing is general-purpose and is not intended as controlled-impedance, delay-matched high-speed differential routing.

JXADC exposes four differential-capable analog pairs (eight signal pins), and the same pins can be used digitally. The XADC itself is a dual 12-bit, 1 MSPS ADC system. For analog joystick wiring, follow UG480 and the Nexys schematic for input mode/range/front-end requirements.

7. Controller and sensor integration options

Option	Interface	Hardware effort	Recommended role
Onboard buttons	GPIO	Very low	First playable prototype / menu / D-pad.
Onboard ADXL362	SPI	Low	Motion/tilt mechanic without extra hardware.
Keyboard / mouse	USB bridge -> PS/2	Low-medium	Desktop control; keyboard movement or mouse aiming.
Pmod JSTK2	SPI	Low	Compact external joystick with digital serial interface.
Analog 2-axis joystick	XADC or external ADC	Medium	True analog stick; validate analog front end.
Pmod NAV / IMU	SPI	Medium	Gyro + accel + magnetometer-based motion control.

Option	Interface	Hardware effort	Recommended role
MCU controller bridge	UART/SPI	Medium	Best way to adapt USB/Bluetooth console-style controllers.
Ethernet input	RMII + network stack	High	Networked control/multiplayer, not first milestone.

8. 2D graphics architecture

The board VGA interface uses four bits per color channel plus horizontal and vertical sync. This gives 4096 simultaneous RGB values. At 640x480, a roughly 25 MHz pixel clock is a conventional target. The FPGA must generate timing and determine the RGB value for every visible pixel.

8.1 Recommended scanline/tile/sprite engine

Stage	Function	Likely storage
VGA timing	x/y counters, HS/VS, active-video window	Registers/LUTs
Background	tile-map lookup -> tile pixel	BRAM tile map + sprite/tile ROM
Sprites	active sprite test, pixel fetch, transparency/priority	BRAM metadata/cache
HUD/text	font ROM + counters	BRAM/ROM
Compositor	priority + RGB444 selection	LUTs/pipeline regs
Output	12 RGB bits + sync	VGA pins

This architecture can avoid a full framebuffer, making 640x480 2D graphics inexpensive in memory. Collision detection can be separate from rendering using tile occupancy, axis-aligned bounding boxes or compact object lists.

8.2 Framebuffer memory math

Resolution	RGB444 color	16-bit depth	Color + depth	On-chip BRAM?
320x240	112.5 KiB	150.0 KiB	262.5 KiB	Yes
640x480	450.0 KiB	600.0 KiB	1,050.0 KiB	No combined
800x600	703.1 KiB	937.5 KiB	1,640.6 KiB	No
1024x768	1,152.0 KiB	1,536.0 KiB	2,688.0 KiB	No

A single 640x480 RGB444 framebuffer fits in the approximately 607.5 KiB of total BRAM, but leaves little space for the rest of the design. A second color buffer does not fit. DDR2 should be considered the normal location for double-buffered framebuffers, depth buffers and sizable textures.

8.3 DDR2 bandwidth perspective

Digilent documents a recommended DDR2 data rate of 650 Mb/s on a 16-bit interface. Multiplying the data rate by the 16-bit width gives roughly 10.4 Gb/s, or 1.3 GB/s raw theoretical transfer bandwidth. Real application bandwidth is

lower because of refresh, row/bank behavior, arbitration and controller efficiency. The VGA reader should use a line FIFO and receive predictable arbitration priority.

9. 3D graphics architecture

A true 3D engine is possible, but the XC7A100T has no hard GPU. The FPGA advantage is that you can build the exact dataflow you need and pipeline it deeply. The cost is development time and careful control of fixed-point precision, memory accesses and timing closure.

9.1 Complexity ladder

Renderer	Relative difficulty	Why it fits / main bottleneck
2.5D raycaster	Low-medium	Small math core; excellent first "3D-looking" game. Texture columns need predictable memory access.
Wireframe	Medium	Transforms + clipping + line rasterization.
Flat/Gouraud triangles	Medium-high	Triangle setup, rasterizer, interpolation, color/depth buffering.
Textured triangles	High	Texture address generation + bandwidth; perspective correction adds arithmetic.
Feature-rich GPU-like pipeline	Very high	Resource/bandwidth/timing complexity grows quickly.

9.2 Suggested fixed-point 3D pipeline

1. Vertex/mesh fetch from BRAM/DDR2
2. Model-view-projection transform using DSP48E1 MACs
3. Clipping and back-face culling
4. Triangle setup and edge/barycentric coefficients
5. Rasterization at 320x240 internal resolution
6. 16-bit depth test and optional texture/light lookup
7. Write color/depth to DDR2 or tiled buffers
8. Read/scale to 640x480 VGA

Prefer fixed-point arithmetic, with widths chosen from error analysis. Floating-point IP can be used selectively, but applying it everywhere tends to consume more DSP/LUT resources and makes high-frequency timing closure harder. The 240 DSP48E1 slices are a major asset for transform, interpolation and filtering pipelines.

10. Clocking, CDC and performance planning

Domain	Initial target	Reason
Board reference	100 MHz	Fixed oscillator.
VGA pixel	~25 MHz for 640x480	Video timing. Generate with clocking resources.
Game state	60 Hz update or submultiple	Deterministic gameplay tick.
Renderer/core	Start near 100 MHz	Simplifies timing closure; increase only after reports.
Sensor buses	Protocol-specific	SPI/I2C/UART rates far below core clock.

Domain	Initial target	Reason
DDR2 UI	MIG-defined	Follow generated controller clocking/constraints.

Do not pass multi-bit buses between unrelated clock domains using only per-bit synchronizers. Use handshakes, Gray-coded counters, asynchronous FIFOs or vendor CDC structures as appropriate. Video scanout and DDR access are especially sensitive to underflow and metastability mistakes.

11. Audio subsystem

The onboard mono output accepts a 1-bit FPGA-generated PWM/PDM-like stream and includes analog filtering/amplification. A retro engine can generate square/triangle/noise channels in RTL and mix them into a higher-rate accumulator feeding PWM. Sampled SFX can be fetched from BRAM/DDR2/microSD. For microphone gameplay, a full high-fidelity decoder is unnecessary if only an energy or clap threshold is needed.

12. Recommended complete game architecture

Layer	Blocks
INPUT PLANE	Buttons/switches | ADXL362 SPI | keyboard/mouse PS/2 | Pmod SPI/I2C/UART | optional XADC
CONTROL	Input conditioning -> normalized controls -> game state/FSM or MicroBlaze
RENDER	Tile/sprite 2D OR fixed-point ray/triangle pipeline
MEMORY	BRAM caches/FIFOs <-> DDR2 arbiter <-> frame/depth/textures; microSD/QSPI assets
OUTPUT	VGA scanout FIFO -> RGB444/HS/VS | audio engine -> PWM/PDM
DEBUG	USB-UART telemetry + LEDs/seven-segment + Vivado ILA if desired

A MicroBlaze soft processor is optional. It can make menus, object updates and asset management more software-like while critical pixel/sensor/audio paths remain RTL accelerators. Because MicroBlaze is synthesized into the programmable logic, it still runs on the FPGA, although a course may define "entirely in FPGA" as pure RTL; confirm that interpretation before choosing the architecture.

13. Development roadmap

Milestone	Exit criterion
Milestone 0 - Bring-up	100 MHz clock/reset, buttons/switches, LEDs, UART hello/debug.
Milestone 1 - Video	VGA color bars -> 640x480 timing -> text/HUD.
Milestone 2 - Playable 2D	Tile map, sprites, collisions, button input, score/state.
Milestone 3 - Motion/audio	ADXL362 calibration/filter and PWM sound.
Milestone 4 - External control	Pmod JSTK2 or external IMU/joystick over SPI/I2C/UART.

Milestone	Exit criterion
Milestone 5 - DDR2	MIG controller, arbitration, scanout FIFO, optional double buffer.
Milestone 6 - Advanced graphics	320x240 internal raycaster/3D pipeline, then optimize.
Milestone 7 - Assets/polish	microSD/QSPI asset loader, menus, effects, profiling/timing closure.

14. Main technical risks and mitigations

Risk	Why it matters	Mitigation
DDR contention	Scanout underflow creates visible corruption.	Line FIFOs; deterministic priority; burst reads.
BRAM exhaustion	640x480 color buffer alone uses ~450 KiB.	Tile/sprite scanline rendering or DDR2 buffers.
Long combinational paths	Sprite priority, division, wide transforms reduce Fmax.	Pipeline, staged reductions, DSP use, reciprocal LUTs.
5 V peripherals	Can damage 3.3 V FPGA I/O.	Use 3.3 V modules or proper level shifting.
USB gamepad assumption	Onboard USB HID bridge is not generic host.	Use keyboard/mouse or external MCU bridge.
Raw sensor noise/drift	Poor game feel.	Calibrate, dead-zone, low-pass, clamp and scale.
Analog XADC misuse	ADC range/front-end differs from ordinary digital GPIO.	Follow UG480 + schematic; condition voltage.
Clock-domain errors	Intermittent hard-to-debug failures.	Explicit CDC design and verification.

15. Recommended proof of concept

Implement a 640x480 RGB444 2D game using a scanline tile/sprite renderer, 16x16 or 32x32 tiles, 8-32 active sprites, five-button controls plus optional ADXL362 tilt, tile/AABB collision detection, a small text HUD and PWM audio. Keep assets in BRAM initially. Add Pmod joystick/IMU next. Integrate DDR2 only after the game loop and VGA path are stable. This sequence isolates the hardest subsystems and gives a playable deliverable early.

16. Datasheet/source package index

The accompanying folder contains .url shortcuts so each source opens at the current vendor-hosted document. This avoids embedding stale duplicate copies of vendor manuals and makes it clear which source is authoritative. A game-focused XDC excerpt is also provided; the Digilent master XDC remains the source of truth.

- Digilent Nexys A7 product page: https://digilent.com/shop/nexys-a7-amd-artix-7-fpga-trainer-board-recommended-for-ece-curriculum/
- Nexys A7 reference manual PDF: https://digilent.com/reference/_media/reference/programmable-logic/nexys-a7/nexys-a7_rm.pdf
- Nexys A7 schematic PDF: https://digilent.com/reference/_media/reference/programmable-logic/nexys-a7/nexys-a7-sch.pdf

- AMD DS180 - 7 Series Overview: https://docs.amd.com/v/u/en-US/ds180_7Series_Overview
- AMD DS181 - Artix-7 Data Sheet: https://docs.amd.com/v/u/en-US/ds181_Artix_7_Data_Sheet
- AMD UG471 - SelectIO: https://docs.amd.com/v/u/en-US/ug471_7Series_SelectIO
- AMD UG472 - Clocking: https://docs.amd.com/v/u/en-US/ug472_7Series_Clocking
- AMD UG474 - CLB: https://docs.amd.com/v/u/en-US/ug474_7Series_CLB
- AMD UG475 - Packaging/Pinout: https://docs.amd.com/v/u/en-US/ug475_7Series_Pkg_Pinout
- AMD UG479 - DSP48E1: https://docs.amd.com/v/u/en-US/ug479_7Series_DSP48E1
- AMD UG480 - XADC: https://docs.amd.com/r/en-US/ug480_7Series_XADC
- ADI ADXL362 datasheet: https://www.analog.com/media/en/technical-documentation/data-sheets/adxl362.pdf
- ADI ADT7420 datasheet: https://www.analog.com/media/en/technical-documentation/data-sheets/adt7420.pdf
- FTDI FT2232H documentation: https://ftdichip.com/document/usb-ic-data-sheets/
- Microchip LAN8720A: https://www.microchip.com/en-us/product/LAN8720A
- Digilent Pmod JSTK2: https://digilent.com/shop/pmod-jstk2-two-axis-joystick/
- Digilent Pmod NAV manual: https://digilent.com/reference/_media/reference/pmod/pmodnav/pmod_nav_rm.pdf
- Digilent master XDC: https://github.com/Digilent/digilent-xdc/blob/master/Nexys-A7-100T-Master.xdc

17. Final design conclusions

- A strong 2D game at 640x480 is a realistic and technically well-matched use of this board.
- The onboard ADXL362 gives you motion input immediately; add a gyro/IMU via Pmod if angular-rate sensing is required.
- Use Pmod GPIO/SPI/I2C/UART for external controls; use 3.3 V logic and level shifting where necessary.
- Do not base a controller plan on arbitrary USB gamepads through the onboard HID bridge. Use keyboard/mouse, Pmod joystick, or an MCU adapter.
- For 3D, prioritize fixed-point pipelining and 320x240 internal rendering, with DDR2 for buffers/textures and BRAM for caches/FIFOs.
- Treat timing reports and synthesized utilization as the final performance authority, not marketing-level maximum clock figures.